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RESEARCH PAPER

Comparative analysis of INLIGHT™-labeled enzymatically depolymerized heparin by reverse-phase chromatography and high-performance mass spectrometry

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Abstract Structural characterization of the microheterogeneity of heparin, heparan sulfate, and other glycosaminoglycans is a major analytical challenge. We present the use of a stable isotope-labeled hydrazide tag (INLIGHT™) with high-resolution/accurate mass (HRAM) reverse-phase LC-MS/MS, which was recently introduced for detailed study of *N*-glycan heterogeneity, to characterize heparinase-digested heparin (digHep) products without the use of semi-volatile ion pairing reagents. Using both full scan LC-MS and data-dependent LC-MS/MS, we identified 116 unique digHep species, a feat possible because of INLIGHT™ labeling. Of these, 83 digHep products were structurally identified, including the 12 standard disaccharides as well as 34 tetra- (DP4), 26 hexa- (DP6), 21 octa- (DP8), and 2 decasaccharides (DP10). Each of the 116 digHep species co-eluted with both light and heavy INLIGHT™ tags ($L/H_{avg} = 1.039 \pm 0.163$); thus enhancing confidence in their identification via MS and MS/MS. This work sets the foundation for INLIGHT™-based comparative analyses of different forms of heparin, heparan sulfate, and other GAGs with high quantitative precision using mainstay reverse-phase HRAM LC-MS/MS.

Keywords Mass spectrometry/ICP-MS · Glycosaminoglycan · Heparin · Chemical derivatization

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Introduction

Heparin is a widely used anticoagulant for treating several thrombotic disorders and high-risk patients undergoing surgery [1]. Pharmaceutical grade heparin is extracted from porcine tissues (e.g., lung and intestines), which is used to prepare three types of heparins [2–4] including unfractionated heparin (UFH: $MW_{avg} \approx 15$ kDa), low molecular weight heparin (LMWH: $MW_{avg} \approx 3$–5 kDa), and ultra-low molecular weight heparin (ULMWH: $MW_{avg} \approx 2$–3 kDa). When administered intravenously, these pharmaceutical forms of heparin bind to antithrombin (Kd ≈ 10 nM) resulting in an antithrombotic activity enhancement of ~1000-fold. The heparin-antithrombin complex formation relies on the presence of a specific sulfated pentasaccharide sequence not present in all the chains of heparin that includes a rare *3-O* sulfate modification on a N-sulfoglucosamine (GlcNS) residue [5]. In addition to this unique pentasaccharide sequence, pharmaceutical grade heparin contains additional sequence variations arising from incomplete residue sulfation that results in high microheterogeneity within and between different heparin dosage forms.

Although heparin-based anticoagulants have been used for over 70 years, the exact chemical structure of the pharmaceutical preparations is poorly defined [6, 7]. For example, the Food and Drug Administration (FDA) states that enoxaparin has chains that are 2 to 32 monomers long, of which chains less than <2000 Da are present at <20 %, those between 2000 and 8000 Da are present at 68 %, and those >8000 Da form <18 % proportion [8]. The ambiguity associated with this description combined with the unnatural structures formed during the chemical/enzymatic processing (e.g., 1,6-anhydro-mannose, galactose, and unsaturation) that are not part of these criteria creates significant challenges when comparing trademark LMWH enoxaparin with generics (e.g., dalteparin, tinzaparin, ardeparin, and certoparin) that are now becoming available to the US market [7]. Furthermore, the structural uncertainty between these different forms of heparin and their derivation from natural sources creates safety and efficacy concerns that have proven lethal in extreme cases [7]. Thus, there are pressing needs for accurate glycoanalytical methods that would allow for comprehensive comparative studies of different pharmacological forms of heparins [9].

There is currently no single analytical technique that can characterize the macro and microheterogeneity of heparin. Heparin, a member of the glycosaminoglycan family (GAG), which includes heparan sulfate, chondroitin sulfate, dermatan sulfate, keratan sulfate, and hyaluronic acid [10], contains alternating uronic acid (GlcA/IdoA)-glucosamine (GlcN) or –glucosaminoacetyl (GlcNAc) disaccharide (DS) repeat units that are differentially sulfated at the 2-*O*, 3-*O*, 6-*O*, and 2-*N* positions. If one assumes the average UFH chain contains ~40 DS units, the theoretical number of unique chemical structures can exceed 1 billion. A common first step for characterizing heparin is to depolymerize (DP) the polysaccharide into smaller fragments which are more easily analyzed by mainstay biophysical techniques. HPLC/UHPLC and capillary electrophoresis (CE) [11, 12] are used to separate the complex depolymerized mixture by molecular weight or charge followed by detection with one or more techniques including UV–vis, fluorescence, nuclear magnetic resonance (NMR) [13, 14], and mass spectrometry [15–32]. It is important to note that well-characterized standards for higher molecular weight oligomers (i.e., above dissacharides) do not yet exist making definitive characterization difficult regardless of the glycoanalytical technique. Despite this limitation, structural characterization of lower molecular weight oligomers is possible with NMR when sufficiently pure oligomers are available or with HPLC/UPLC coupled to mass spectrometry (LC-MS/MS) when either separation efficiency or MS/MS efficiency is sufficient to resolve isobaric species. These advanced analyses can provide critical structure-function information related to heparin biology and novel binding partners.

High/ultrahigh-performance liquid chromatography coupled to mass spectrometry (LC-MS or LC-MS/MS) is commonly used to characterize and quantify the end products of enzymatically digested heparin (digHep), which typically includes 12 common DS's as well as incompletely depolymerized heparin oligomers such as tetra- (DP4), hexa- (DP6), octa- (DP8), etc. The vast majority of LC-MS methods use either reverse phase ion pairing (RP-IP) [16, 18, 22, 26, 32–35] or hydrophilic liquid chromatography (HILIC) [31, 36, 37] to separate the highly anionic sulfated digHep species. RP-IP LC-MS/MS, which uses semi-volatile ion pairing agents (e.g., tetraethylamine), provides the highest level of separation efficiency by shielding the negatively charged sulfate groups and enhancing interaction with the reverse phase solid support [34]. Although effective, RP-IP severely contaminates HPLC/UHPLC systems and fouls ESI sources with the semi-volatile alkyl amines rendering instruments less sensitive over the long-term [38].

An alternative approach to RP-IP is chemical derivatization, where digHep species are rendered more hydrophobic and therefore retained on standard reverse phase columns. Two main types of labeling strategies have been developed in this context: (1) permethylation [27, 39–41] and (2) reducing end chemical tags [19, 30, 42–49]. The recently introduced permethylation methods for GAGs introduce a methyl group on the free hydroxyl and amine groups followed by desulfation and perdeuteroacetylation. This approach shows great promise for sequencing sulfated GAGs but the method requires additional refinement to overcome issues related to reaction efficiency and solubility.

Functionalizing the reducing end (RE) of glycans/GAGs with a chromophore or stable isotope labeled tag has been broadly applied to structural/compositional analysis [43, 45, 50]. Several chemical tags have been introduced already with a variety of linking chemistries (e.g., reductive amination [19], hydrazide [46–49, 51], oxime [52]) and physiochemical properties (e.g., ionic, neutral, affinity, etc.) to span multiple glycoanalytical applications. Hydrazide linking chemistry is an attractive strategy for coupling hydrophobic tags with stable isotope labels to the RE of glycans [45]. The tagging chemistry does not introduce unwanted salts, is relatively fast (<3 h), and can be immediately injected onto LC-MS/MS without clean-up. Recent developments with this class of RE functionalization have focused almost entirely on N-glycans released from proteins [46–49, 51]. However, extending this technology for GAG analysis could also be potentially rewarding.

In this study, we characterized the final products of heparinase-digested UFH (digHep) using a combination of stable isotope labeled (SIL) hydrazide (INLIGHT™) tags [47] and high resolution accurate mass (HRAM) LC-MS/

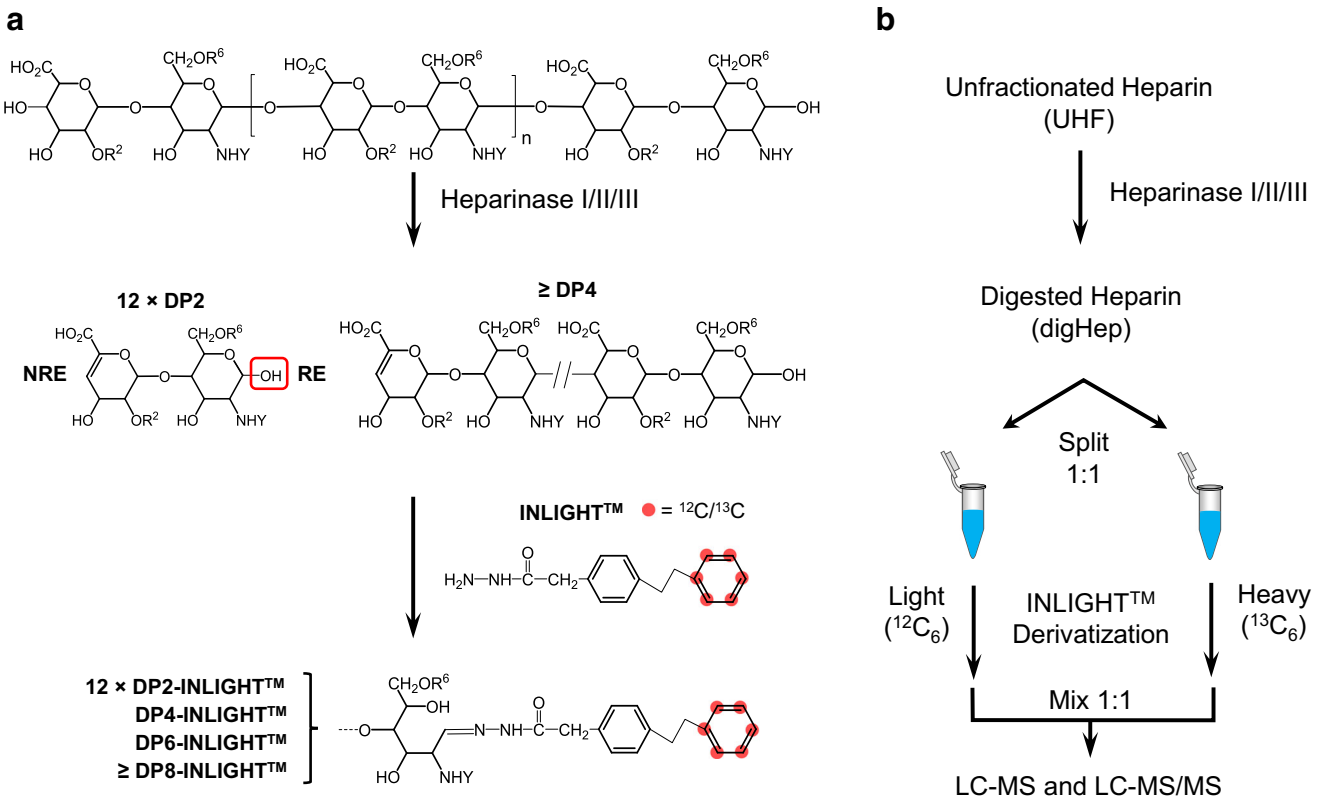


Fig. 1 **a** General chemical structure of the linear polysaccharide heparin showing alternating glucuronic acid-glucosamine disaccharide repeat units. The Y group can exist as an H, C2H3O, or SO3; R2 and R6 can exist as an H or SO3; and the average heparin chain has $n = 40–50$ disaccharide repeat units. Enzymatic digestion of heparin (digHep) with heparinase I/II/III primarily produces 12 disaccharide species (DP2) and incompletely depolymerized oligomers (≥DP4) which contain an unsaturated glucuronic acid 4,5 double bond and retains the reactive reducing ends (RE) which can be reacted with the light or heavy INLIGHTTM tag via the hydrazide linkage. **b** Experimental overview with the digHep sample being split into equal fractions followed by light or heavy labeling, 1:1 mixing, and either full scan LC-MS or data-dependent LC-MS/MS analysis. *NRE* = non-reducing end; *RE* = reducing end

MS. Our interest in using the INLIGHT™ tagging approach for characterizing digHep was based on several aspects unique to highly sulfated GAGs. (1) Separation modalities for LC-MS/MS of un-derivatized digHep have been restricted to HILIC or RP-IP HPLC that use semi-volatile ion pairing reagents (e.g., tetraethylamine) that foul ion optics; (2) accurate quantification of digHep products on a global scale is limited by a dearth of purified standards; (3) poor ionization efficiencies of highly hydrophilic un-derivatized digHep; and (4) simultaneous detection of each light and heavy INLIGHT™ labeled fragments separated by 6.0201 m/z ±5 ppm raise the confidence of identifying digHep species. Following SIL INLIGHT™ labeling of a split (i.e., identical) digHep sample, we were able to identify 116 unique digHep species with light-heavy SIL pairs ranging from the 12 standard DP2 species up to 2.7 kDa decameric oligomers (DP10). The ability to precisely quantify 116 unique digHep species using standard reverse-phase LC-MS/MS provides a clear path forward to better defining pharmacological forms of heparins (e.g., UFH, LMWH, and ULMWH) and elucidate their similarities and differences between manufacturers and within manufactured batches.

Experimental

Chemical reagents INLIGHT™ Glycan Tagging kit was purchased from Cambridge Isotope Laboratories. Ammonium acetate (HPLC-grade; ≥99.0 %) and acetic acid (LCMS grade) were purchased from Sigma-Aldrich. HPLC-grade water, acetonitrile, and methanol were purchased from Burdick and Jackson.

Depolymerized UFH preparation and INLIGHT™ derivatization Porcine UFH was purchased from Sigma-Aldrich and used as received. Heparinases I, II, and III were gifts from Dr. Kuberan Balagurunathan (University of Utah, Salt Lake City, UT). A mixture of heparinases I, II, and III (1 mIU/μl each, 10 μl total volume) was added to phosphate-buffered saline (10 mM sodium phosphate, 137 mM NaCl, and 2.7 mM KCl), pH 7.4, solution of UFH (400 μg) such that the total volume was 250 μl. The enzymatic reaction was allowed to continue for 48 h at 37 °C and then stored at −80 °C. The concentration of the digested heparin (digHep) was calculated from the absorbance at 232 nm and found to be 1.6 μg/μl.

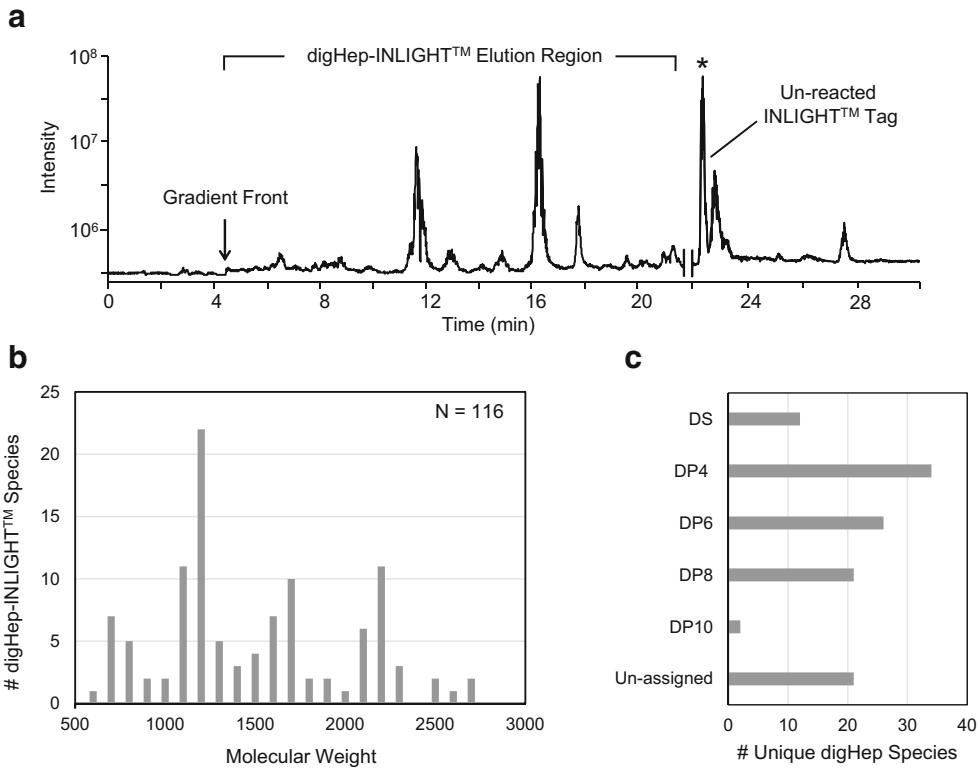


Fig. 2 **a** LC-MS total ion chromatogram of differentially INLIGHT-tagged digHep. The elution region spans 4–22 min with the corresponding intensities given on the *y*-axis. The eluting species from 22 to 30 min are plotted at 0.25× the *y*-axis scale where the most abundant peak (*asterisk*) intensity = 3.17 × 108. **b** Number of unique INLIGHTTM-tagged digHep species as a function of molecular weight. Unique species requires a light/heavy pair and does not include species with sodiated adducts or sulfate loss. **c** Number of species classified as disaccharide (DP2), tetrameric (DP4), hexameric (DP6), octameric (DP8), decameric (DP10), and unassigned digHep oligomers

A 50-μl aliquot of the digHep solution (80 μg) was split into two equivalent 25 μl aliquots and dried in a Speedvac at 50 °C for 1 h. The two dried digHep samples were then reconstituted in 2 × 160 μl solutions of acetic acid/water/methanol (12.5/12.5/75; vol/vol/vol) containing either 1 mg/ml of light ($^{12}C_6$) or 1 mg/ml of heavy ($^{13}C_6$) INLIGHT™ tag. The two solutions were vortexed and then incubated for 3 h at 56 °C. The solutions were then mixed 1:1 (vol/vol), dried in a Speedvac at 50 °C for 2 h, reconstituted in 200 μl of mobile phase A, vigorously vortexed for 1 min, and then centrifuged at 15,000 rpm for 5 min. Approximately 180 μl of solution was removed from the top being careful not to disturb the pellet of un-reacted INLIGHT™ tag at the bottom of the vial. Five microliter volumes of the resulting 180 μl solution were analyzed by LC-MS and LC-MS/MS.

LC-MS/MS system The high-performance LC-MS/MS system consisted of an Eksigent 415 NanoFlex HPLC (Sciex) fitted with the 1–10-μl/min gradient flow module, an Acclaim (ThermoFisher Scientific) PepMap RSLC (300 μm × 15 cm, 3 μm, 100 Å) C-18 HPLC reverse phase column, and a Q-Exactive tandem Orbitrap mass spectrometer (ThermoFisher Scientific). Mobile phase A was 20 mM ammonium acetate/acetonitrile (98/2; vol/vol) and mobile phase B was 20 mM ammonium acetate/acetonitrile (40/60; vol/vol). The flow rate and column temperature were kept constant at 6 μl/min and room temperature, respectively. The 35-min HPLC gradient was as follows: 0–10 min (50 % B), 10–20 min (50–98 % B), 20–30 min (98 %), 30–32 min (98–50 % B), and 32–35 min (50 %).

The Q Exactive HESI source was operated in negative ion mode using the following parameters: ESI probe = −2.5 kV, capillary temperature = 250 °C, sheath gas = 5, S-lens RF = 62. The INLIGHT™-digHep sample was analyzed sequentially by the Q Exactive using a full scan MS method (LC-MS) and then a data-dependent method (LC-MS/MS) both of which used identical aforementioned HPLC and HESI conditions. The full scan method consisted of the following parameters: *m/z* range = 310–1500 *m/z*; 70 k RP, AGC = 3E6, Max IT = 250 ms, and microscans = 1. The data-dependent method consisted of the following MS parameters: *m/z* range = 310–1500 *m/z*; 70 k RP, AGC = 1E6, Max IT = 250 ms, microscans = 1 and MS/MS parameters: 17.5 k RP, AGC = 1E5, Max IT = 250 ms, # of MS/MS = 12 (i.e., top-12), isolation = 2 *m/z*, NCE = 26, and a dynamic exclude = 5 s. The Q Exactive was calibrated immediately prior to the LC-MS and LC-MS/MS analyses using the ESI negative ion calibration solution (ThermoFisher Scientific).

Results

Heparin digested with heparinase I/II/III produces a complex mixture of 12 disaccharides (DP2) and oligomers including tetra- (DP4), hexa- (DP6), octa- (DP8), decasaccharides (≥DP10) and others depending on the extent of digestion

(Fig. 1a). The resulting digestion products contain a double bond (Δ4,5) at the non-reducing end (NRE) and a reactive hydroxyl group at the RE, which was functionalized with the INLIGHT™ hydrazide tag in this study. The final INLIGHT™-labeled digHep structure is given in Fig. 1a, which is representative of the digHep species (i.e., DP2, and others). The experimental design for this study that uses the general reaction scheme in Fig. 1a is shown in Fig. 1b. The digHep sample was split into two equivalent fractions and then labeled with either light-$^{12}C_6$ [L] or heavy-$^{13}C_6$ [H] INLIGHT™ hydrazide tagging reagent for 3 h at 56 °C under acidic conditions. The solutions were then combined in equimolar proportion and analyzed by reverse phase LC-MS and LC-MS/MS.

The light/heavy labeled digHep samples described in Fig. 1 provide three critical elements in our efforts to develop a high resolution/mass accuracy (HRAM) reverse phase LC-MS/MS platform for comparative analysis of heparins: (1) the number of unique INLIGHT™ labeled digHep species that can be identified and quantified using the HRAM L/H pair (Δ6.0190 Da), (2) the effect of the INLIGHT™ tag on the digHep separation on a standard reverse phase LC system, and (3) the use of L/H-labeled MS/MS spectra to identify non-reducing (i.e., no Δ6.0201 Da shift) and reducing (i.e., Δ6.0201 Da shift) end fragments. The L/H INLIGHT™-labeled digHep sample was initially analyzed via high resolving power (70 k RP) full scan LC-MS to generate a high scan density across the eluting peaks. An initial L/H peak list was generated and isobaric species that were chromatographically separated were considered unique. The peak lists were then compared with theoretical digHep structures and their respective molecular weights to reduce the number of potential candidates and preliminary identification. Following the LC-MS run, the sample was re-analyzed using the identical LC gradient but with a Top-10 data-dependent LC-MS/MS method. The LC-MS/MS run confirmed the identification of all 12 heparin/heparan sulfate DP2 species and provided structural information for some DP oligomers.

The full scan LC-MS total ion chromatogram (TIC) for the L/H (1:1) digHep-INLIGHT™ sample is shown in Fig. 2a. Labeled species with co-eluting L/H pairs were detected from 4 to 22 min (shown at full intensity scale) and un-reacted INLIGHT™ tag eluted at 22.5 min (shown at 0.25× intensity scale). The LC-MS dataset was interpreted manually in XCalibur™ by searching for predicted species based on theoretical *m/z* and visually detecting distinct L/H pairs. Within the 4–22-min elution region, a total number of 116 unique INLIGHT™-labeled digHep species were detected spanning a molecular weight range of 500–2800 Da (Fig. 2b). Isobaric L/H species that were chromatographically separated by 30 s at $FWHM_{50}$ were identified as unique. The 116 unique species included all 12 standard heparin/heparan sulfate DP2; 34 × DP4, 26 × DP6, 21 × DP8, 2 × DP10, and 21 unassigned (Fig. 2c). The unassigned 21 species had co-eluting L/H pairs but did not align with theoretical *m/z* values from our in house database. This unassigned species may contain unusual or rare modifications including 3-*O*-sulfation and anhydro ring formation, which will require development of a comprehensive database and additional chromatographic/mass spectrometric experiments. The average L/H ratio for all 116 unique INLIGHT™-tagged depolymerized heparin species was 1.039 ± 0.163 (Electronic Supplementary Material (ESM) Fig. S1).

The quantification of 12 DP2 repeat units produced after heparinase I/II/III digestion is commonly used as a proxy for measuring the extent and type of sulfation in the original heparin/heparan sulfate sample. All of the 12 DP2 repeat units from the digHep sample were detected in the full scan LC-MS dataset. Extracted ion chromatograms (XICs) for all 12 $^{12}C_6$-INLIGHT™-labeled DP2 are shown in Fig. 3 including three sets of isobaric species: II-A and III-A ($[M\text{-}H^+]^{1-}_{Theoretical} = 694.1923$); I-H, II-S, and III-S ($[M\text{-}H^+]^{1-}_{Theoretical} = 732.1386$); and IV-S, II-H, and III-H ($[M\text{-}H^+]^{1-}_{Theoretical} = 652.1818$). Each set of isobaric DP2 species was chromatographically resolved and showed the corresponding L/H (1:1) INLIGHT™-labeled

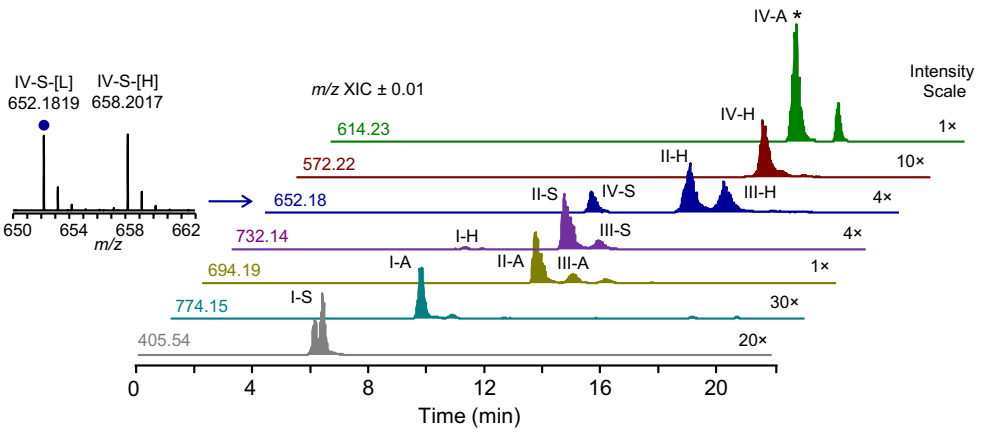
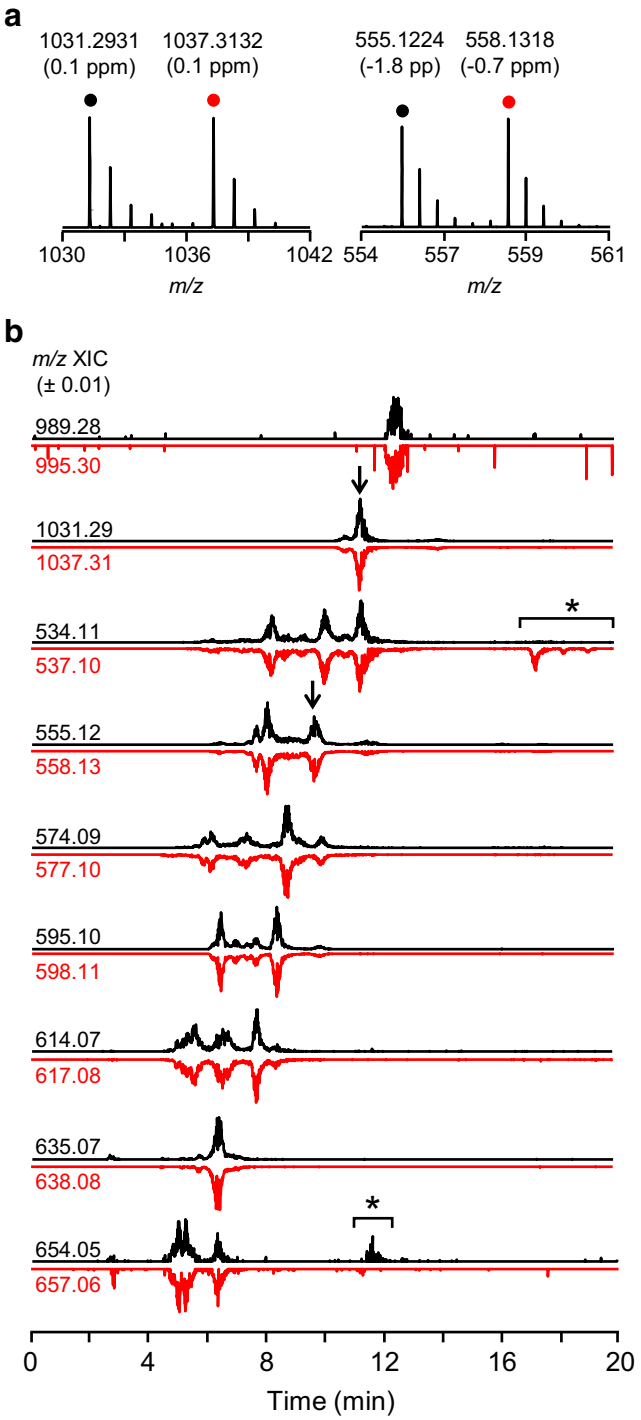


Fig. 3 Extracted ion chromatograms for all 12 12C6-INLIGHTTM-tagged disaccharide species from the full scan LC-MS. The 12 13C6-INLIGHTTM-tagged DP2 co-eluted at the same intensity but the overlayed XICs are not shown for simplicity. A representative L/H isotopic distribution pair for IV-S-INLIGHTTM is shown in the inset. *Asterisk*, intensity = 2.05 × 107



◀ **Fig. 4** **a** Representative isotopic distributions from the full-scan LC-MS chromatogram for two unique DP4 species labeled with light (*black dot*) and heavy (*red dot*) INLIGHTTM tags. Monoisotopic mass-to-charge labels are given above each light and heavy species with respective mass measurement accuracies given in parenthesize (parts-per-million; ppm). **b** Extracted ion chromatograms for all light/heavy labeled DP4 species detected in the full-scan LC-MS analysis of INLIGHTTM labeled digHep. *Black* and *red traces* represent light and heavy labeled DP4 species, respectively. The two arrows above the 1031.29 and 555.12 indicate the locations for the two representative isotopic distributions shown above. The *asterisk* indicates regions where peaks were extracted for one tag (i.e., light or heavy) but not the other indicating contaminant species

pairs with the representative L/H IV-S mass spectrum given in the inset of Fig. 3.

Partially digested heparin products were also observed from the full scan LC-MS dataset including DP4, DP6, DP8, and DP10. Light/heavy pairs with *m/z* corresponding to nine possible DP4 masses were detected. Representative L/H isotopic distributions (1:1) for 2 of the 9 are shown in Fig. 4a for $[M\text{-}H^+]^{1-}{}_{Theoretical} = 1031.2932/1037.3133$ and $[M\text{-}2H^+]^{2-}{}_{Theoretical} = 555.1214/558.1314$. Figure 4b shows the XICs

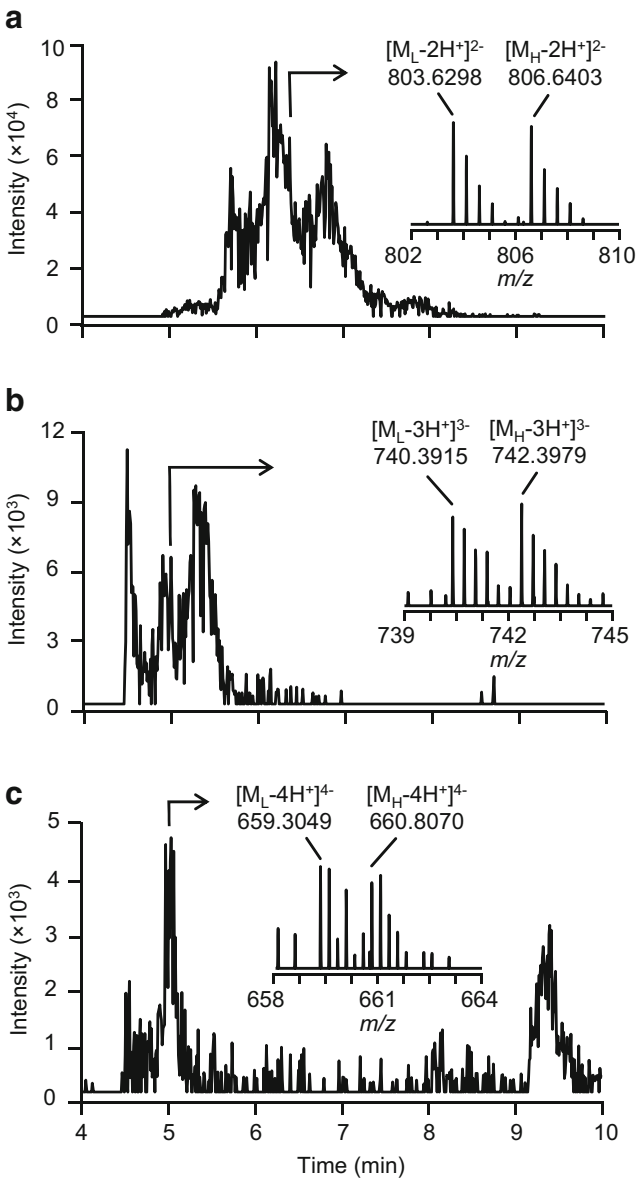


Fig. 5 Extracted ion chromatograms and representative light/heavy isotopic distributions from full-scan (70 k RP) LC-MS analysis showing a **a** DP6; **b** DP8; and **c** DP10 species. ML and MH represent depolymerized heparin oligomers labeled with either light or heavy INLIGHTTM tags, respectively

for the 9 L/H (black/red) DP4 pairs including black arrows indicating the location for the representative isotopic distributions in Fig. 4a that were collected. It is interesting to note that select peaks were observed containing only one tag (either light or heavy) suggesting the presence of contaminants (labeled with an asterisk). This bodes well for performing high-fidelity quantitative characterization of how two heparin samples (e.g., reference vs. test) could differentially respond to a set of enzymatic digestions.

Elution order correlated with the degree of sulfation ranging from high sulfation for 654.05/657.06 DP4 species ($5 \times SO_3$) from 4 to 7 min to low sulfation for 989.28/995.30 DP4 species ($1 \times SO_3$) at 12.5 min. A total of 49 high molecular weight DP6, DP8, and DP10 oligomers with L/H isotopic pairs were detected in the LC-MS datasets. Importantly, these species do not include sodiated adducts, which were present but not identified. The intensity of monosodiated peaks of the light labeled disaccharides and light DP4 species was determined and compared with the nonsodiated species to obtain a relative percentage of sodiated/nonsodiated (NaL/L) within the heparin digest over the length of the chromatographic run. The average percentage of NaL/L was found to be 15.3 % for 27 disaccharide and DP4 species identified with mass accuracies between −0.26 and 0.51 ppm. Representative XICs ($^{12}C_6$-INLIGHT™-labeled species) and isotopic distributions for select nonsodiated DP6, DP8, and DP10 monoisotopic masses are shown in Fig. 5. The XICs for $3 \times$ DP6 ($MW = 1609.2719$), $\sim 2 \times$ DP8 ($MW = 2224.1955$), and $1 \times$ DP10 ($MW = 2641.2476$) are shown in Fig. 5a, b, and c, along with a representative light/heavy pair isotopic distributions, respectively. Out of the 24 DP6 species detected, 13 distinct m/z species were identified including nine sets of isobaric species that were chromatographically separated. A total of 21 unique DP8 species associated with nine distinct m/z values were detected. The two detected DP10 species eluted immediately after the gradient front indicating virtually no retention on the reverse phase column. However, the two species had corresponding L/H monoisotopic pairs and unique molecular weights. Overall, the full scan LC-MS method provides respectable coverage of digHep species with the inherent capability of relative quantitation for comparative analysis of heparin/heparan sulfate.

Immediately following the full scan LC-MS analysis of the digHep-INLIGHT™ (L/H, 1:1) sample, the sample was analyzed again using the same LC method but in a top-12 data-dependent MS/MS mode to sequence the ~116 unique digHep species. Figure 6a shows the resulting XIC for $^{12}C_6$-INLIGHT™-labeled I-S (405.54) with full scan (70 k RP) and MS/MS events (17.5 k RP) for the light $^{12}C_6$-INLIGHT™-labeled ($m/z = 405.54$) and heavy $^{13}C_6$-INLIGHT™-labeled I-S ($m/z = 405.55$) isotopes. The isotopic distributions from 405 to 411 m/z for the full scan MS in Fig. 6a is shown in Fig. 6b. Excellent MS/MS fragmentation spectra from the INLIGHT™-labeled I-S structure (Fig. 6c)

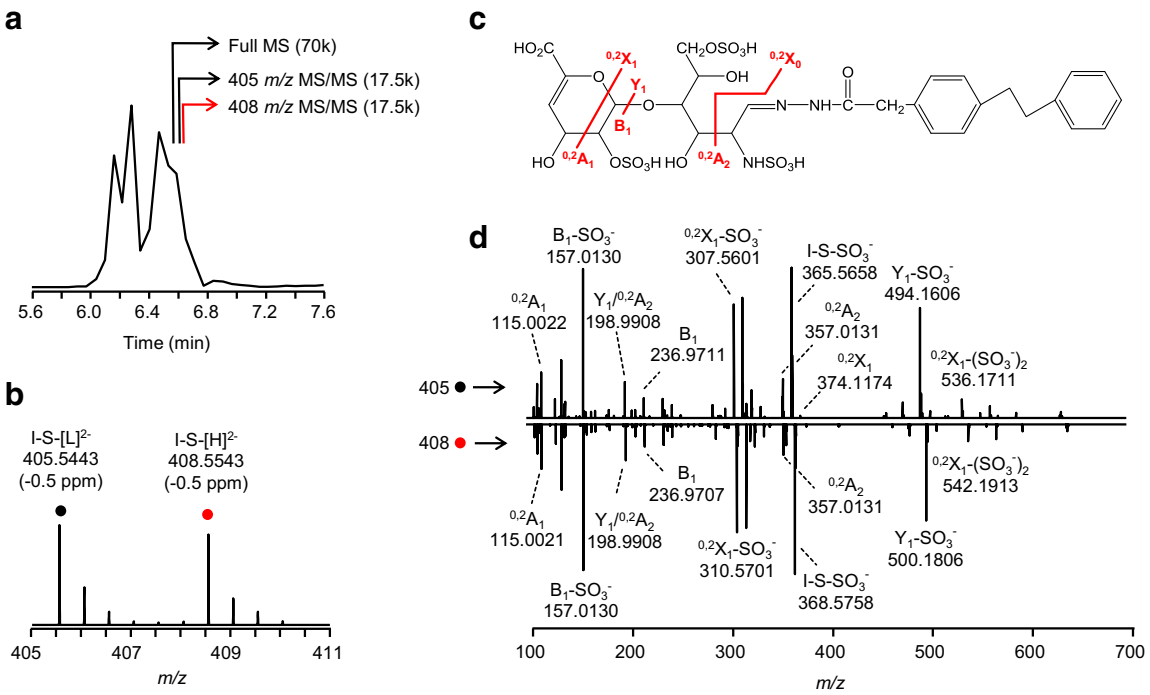


Fig. 6 **a** LC-MS/MS extracted ion chromatogram for 405.54 m/z which corresponds to 12C6-INLIGHTTM-tagged I-S. **b** Full scan MS of light/heavy I-S pairs with labeled monoisotopic m/z and resulting mass measurement accuracies (ppm) given in *parentheses*. Both light (*black dot*) and heavy (*red dot*) were selected for MS/MS to fragment the two 13C6-INLIGHTTM-tagged I-S species. **c** Fragmentation patterns for both monoisotopic species and **d** corresponding tandem mass spectra are shown

were observed for both light and heavy forms (Fig. 6d). Prominent I-S Domon-Costello [53] fragments $^{0,2}A_1$, $^{0,2}X_1$, Y_1-SO_3, B_1, $^{0,2}A_2$, $^{0,2}X_0$, and $Y_1/^{0,2}A_2$ were observed in agreement with a previous study of underivatized heparin/heparan sulfate DP2 [20]. Importantly, the light and heavy tags give a 6.0201-Da mass shift for X, Y, and Z-type ions (reducing end fragments) providing a valuable signature for identifying fragments in higher molecular weight depolymerized oligomers. For the example shown in Fig. 6, the L/H fragments for $^{0,2}X_1$-SO_3, $^{0,2}X_1$-$(SO_3)_2$, and Y_1-SO_3 gave *m/z* shifts of 3.010 (2-), 6.0200, and 6.0202, respectively. A summary of detected $^{12}C_6$-INLIGHT™-labeled fragment ions for all 12 heparin/heparan sulfate DP2 species is given in Table 1. Additional Domon-Costello fragments were observed in these MS/MS spectra such as loss of one or more SO_3, H_2O, and CO_2 but were not included due to space limitations. Although the experimental design used in this study was limited to equimolar proportions for the purpose of mapping the digHep sample, it is clear from the data that comparative labeling of varying types of digHep samples (or digested heparan sulfate) with INLIGHT™ would provide a powerful means for measuring the relative differences in 12 DP2 species as well as relative differences for incomplete digestion products. Such exhaustive analysis (e.g., full vs. partial digestion) may be able to detect more digested species than that being reported here.

Finally, to highlight the capacity of this novel approach to obtain both detailed structural and quantitative information from larger molecular weight species, representative MS/MS spectra for a light and heavy labeled DP4 are shown in Fig. 7. Extensive sequence information was obtained for the DP4 species (Fig. 7a) that allowed for the confident identification of the DP4 as a IIH-IVA (Δ4,5UA-GlcN-UA-GlcNAc-INLIGHT™) sequence. Nearly all NRE fragments such as $^{0,2}A_1$, B_1, C_1, $^{0,2}A_2$, C_2, $^{0,2}A_3$, $^{0,2}A_4$, and $^{0,1}A_4$ and L- or H-labeled RE fragments such as $^{0,2}X_3$, $^{0,3}X_0$, $^{0,2}X_1$, Y_3, and Y_3-SO_3 were readily detected. Each RE fragment contained the reproducible 6.0201-Da mass shift allowing quick identification. Interestingly, N-N bond breakage within the hydrazide tag (T_1) was observed providing another signature ion for downstream identifications in complex LC-MS/MS datasets.

Discussion

The present study identified 116 unique enzymatically digested heparin (digHep) species using the newly introduced INLIGHT™ hydrazide tag and standard reverse phase LC-MS/MS fitted with a commercial 300 μm × 150 mm microflow reverse phase HPLC column. The unique species included all 12 standard heparin/heparan sulfate disaccharides, 34 × DP4, 26 × DP6, 21 × DP8, 2 × DP10, and 21 unassigned digHep species. The use of stable isotope labeled

Table 1 Measured intact (DP2-[L]) and subsequent fragment ions (normalized collision energy = 26) detected in the data-dependent LC-MS/MS analysis of INLIGHTTM-tagged digHep sample

	INLIGHTTM-labeled (light [L]-12C6) disaccharide MS/MS fragment ions										
	DP2-[L]	0,2A1	B1	C1	0,2A2	0,2X1	Y1	Z1	0,2X0	Y1/0,2A2	
I-A	774.1505 (−1.7)	115.0021 (13.9)	236.9692 (7.9)				536.1724 (−2.9)	518.1606 (−0.8)	336.1722 (−1.3)	198.9910 (4.0)	
II-A	694.1929 (−0.9)	115.0018 (16.5)	157.0131 (7.1)	175.0237 (6.3)	357.0135 (−0.6)	578.1812 (0.4)	536.1716 (−1.5)	518.1602 (0.0)	336.1729 (−3.3)		
III-A	694.1931 (−1.2)		236.9712 (−0.6)		357.0138 (−1.4)	578.1797 (2.9)			336.1727 (−2.7)		
IV-A	614.2357 (−0.3)	115.0021 (13.9)	157.0130 (7.6)	175.0238 (5.7)					336.1715 (0.9)		
I-H	365.5659 (−0.5)	115.0022 (13.0)					494.1609 (−1.3)	477.1357 (45.7)		198.9912 (3.0)	
II-H	652.1828 (−1.5)	115.0019 (15.7)	157.0130 (7.6)	175.0238 (5.7)	357.0140 (−2.0)	536.1716 (−1.4)	494.1609 (−1.3)	477.1342 (48.9)	294.1612 (−0.1)	198.9909 (4.5)	
III-H	652.1819 (−0.2)		236.9708 (1.1)		357.0134 (−0.3)	536.1730 (−4.1)					
IV-H	572.2252 (−0.3)	115.0020 (14.8)	157.0129 (8.3)	175.0238 (5.7)							
I-S	405.5442 (−0.5)	115.0022 (13.0)	236.9711 (−0.2)		357.0131 (0.6)	374.1174 (1.6)				198.9908 (5.0)	
II-S	365.5653 (1.1)	115.0021 (13.9)	157.0130 (7.6)		357.0536 (−112.9)	307.5600 (0.6)				198.9907 (4.0)	
III-S	365.5652 (1.1)	115.0023 (12.2)				307.5615 (−4.3)	494.1597 (1.1)	477.1350 (47.2)			
IV-S	652.1819 (−0.2)	115.0022 (13.0)	157.0142 (0.0)	175.0239 (5.1)		536.1730 (−4.1)	494.1649 (−9.4)				

Experimental *m/z* values for each fragment and subsequent mass measurement accuracy (ppm) are shown

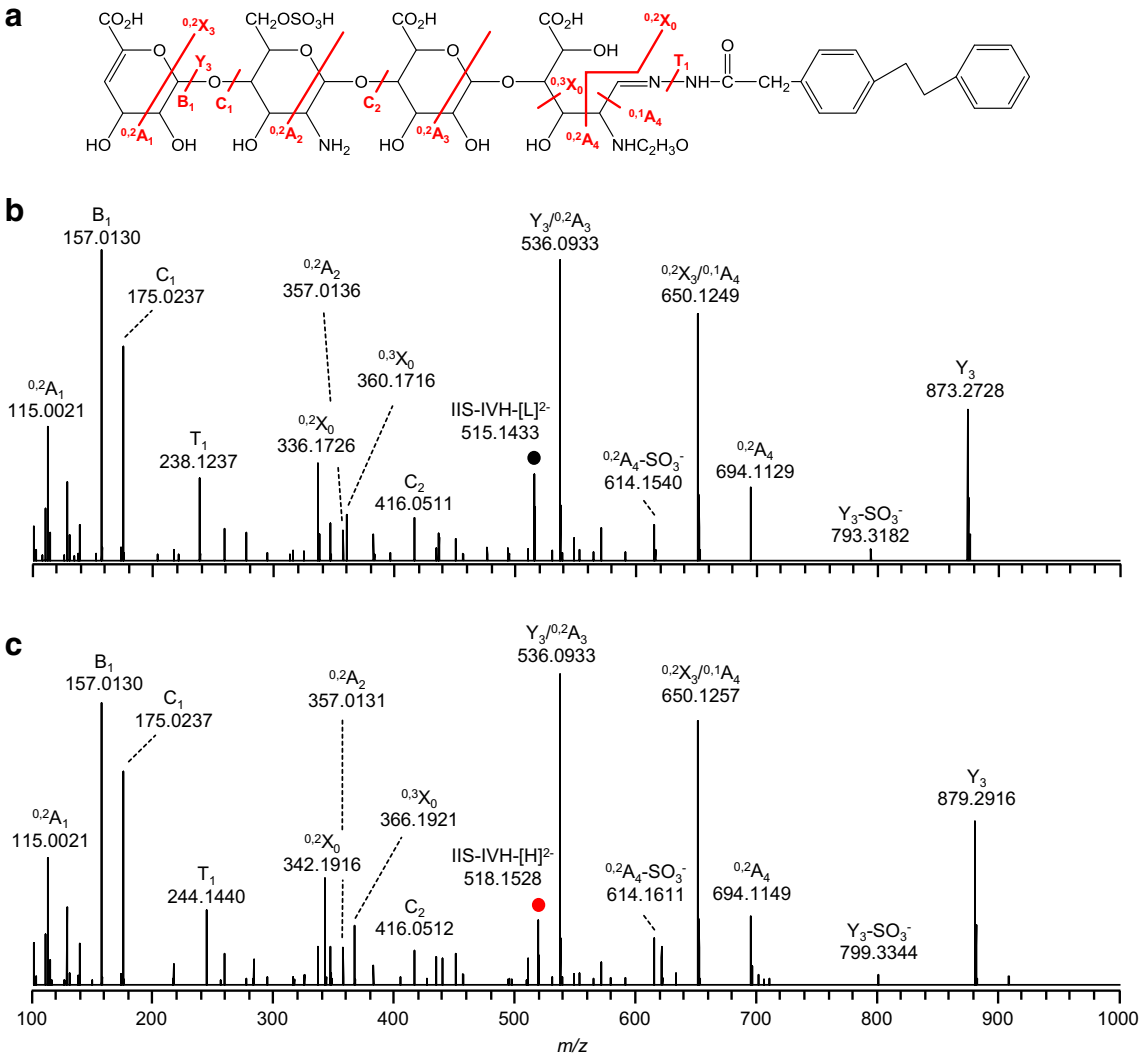


Fig. 7 **a** Molecular structure for IIH-IV-A-INLIGHTTM and labeled fragment ions identified by LC-MS/MS. MS/MS spectra for **b** light and **C** heavy INLIGHTTM tagged IIH-IV-A DP4

INLIGHT™ hydrazide tagging provided a hydrophobic tag that improved retention on the reverse phase HPLC column while also producing light and heavy pairs for structural identification (MS and MS/MS) and relative quantification. The work presented herein provides a platform for globally characterizing and quantifying enzymatically digested heparins (UFH, LMWHs, ULMWHs) spanning the basic disaccharide repeat units up to incompletely digested oligomers (e.g., DP4, DP6, DP8, etc.).

Mass spectral sequencing is a longstanding goal for researchers to assess microheterogenous differences between pharmaceutical preparations of heparins. A more global value of GAG sequencing is to correlate biological phenotype with structure. However, these highly sulfated linear polysaccharides present enormous analytical challenges that have only been partially met with innovations in separation science, chemical derivatization, and HPLC/UHPLC and MS/MS technology.

In this work, we have introduced the use of stable isotope labeled hydrazide tag (INLIGHT™) for analysis of digested heparin species which was originally developed for N-glycan analysis [46–49]. The tag was empirically designed to rapidly react at the reducing ends of N-glycans with little post-derivatization clean-up. The tag facilitated improved N-glycan separation efficiency, enhanced ionization efficiency by introducing a hydrophobic tail on the N-glycan, and allowed for the introduction of stable isotopes to facilitate relative quantification by LC-MS.

The present work extends the use of this tag toward sulfated linear polysaccharides, which represent some of the most hydrophilic and highly anionic biomolecules in nature. This physiochemical property has rendered these biomolecules difficult to separate by traditional chromatographic modalities, particularly reverse phase chromatography which makes up the majority of LC-MS/MS instrument configurations. The use of semi-volatile ion pairing reagents (e.g., triethylamine-

TEA) dramatically improves the separation of sulfated GAGs but at the cost of irreversibly contaminating HPLC systems and fouling ESI sources and MS optics. Our long-term goal is to develop a routine method for characterizing and quantifying sulfated GAGs using standard reverse phase HPLC chromatography coupled to HRAM MS/MS systems.

The current study on INLIGHT™ tagging of enzymatically digested heparin with standard reverse phase liquid chromatography coupled to HRAM MS/MS shows promise in overcoming some of the traditional barriers to GAG analysis. A mock comparative digHep sample labeled with light and heavy INLIGHT™ tags yielded 116 unique disaccharide and incompletely digested oligomers via LC-MS and LC-MS/MS analysis. Identifying unique digHep species is very challenging due to the increase in isobaric species with increasing molecular weight. The co-eluting light and heavy digHep pairs with 6.0201 Da shifts makes it easier to identify digHep species versus non-digHep species with single isotopic distributions. Furthermore, the dual MS/MS spectra of the light and heavy monoisotopic peaks provides an easy signature to separate NRE vs. RE fragment ions, particularly with increasing oligomer molecular weight. These factors are of great importance due to the fact that user-friendly and widely applicable software for interpreting LC-MS/MS GAG datasets is not yet available. With the expansion of our spectral libraries and experimentally determined fragment ions, we can begin to use this approach for comparative analysis of heparins (UFH, LMWHs, and ULMWHs) from different sources and identify sequences with novel biological function(s). In fact, the success of this approach for heparins suggests that sulfated GAGs, e.g., chondroitin sulfate, dermatan sulfate, and keratan sulfate, could also be possibly characterized in detail using INLIGHT technology.

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Compliance with ethical standards

Conflict of interest The authors do not have any conflict of interests to declare.

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